GGUF quantization-1

1) Basic idea

llamaCPP : llamaCPP is llm inference framework same as PyTorch

GGUF: a binary format that is optimized for quick loading and saving of models, making it highly efficient for inference purposes. offers a wide array of state-of-the-art integer quantization techniques (2-bit to 8-bit)

Why llamacpp over PyTorch:
minimal dependencies, superior quantization support, and high performance on a wide range of consumer hardware(cross-platform optimized : CPU{arm,x86}, GPU{amd , nvidea, apple, silicon}), including CPU-only setups and integrated GPUs. PyTorch is a powerful deep learning framework primarily designed for research and training, whereas llama.cpp is a specialized, efficient inference engine

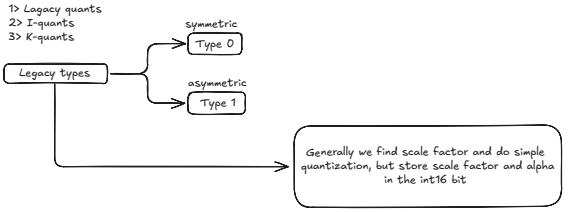
GGUF developed by : 🌐 https://github.com/ggerganov

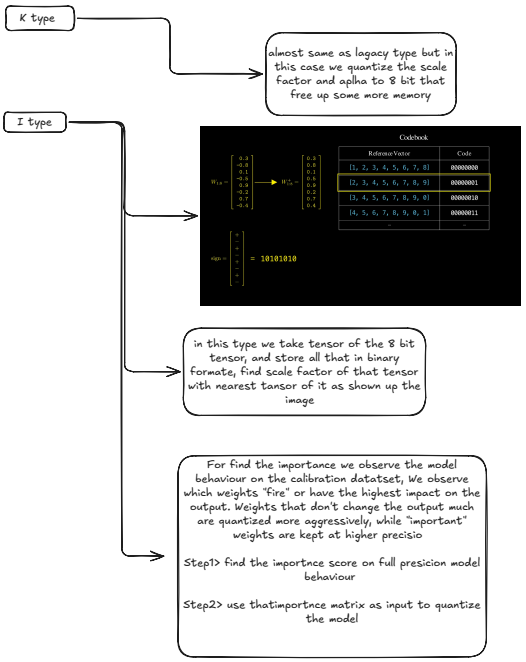
2)basic steps

1> Install llamacpp
2> use llama cpp to convert your model to the gguf formate
3> run llama quantize to shrink the model
4> run model inference with command llama cli -m model...

3)Quantization algorithams used in llamacpp

1> Lagacy quants
2> I-quants
3> K-quants

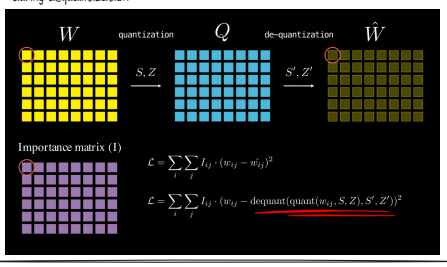




The Importance Matrix (can be used with any of three)

The iamge explain that how we able to retrive clossest W for the important weight where we find loss base on the importance

where minimum $\mathcal{L}$ lead to get maximum precision during dequantization



its just basic overview of the GGUF , maybe in future i am gonna publish about it even with practical notebook code and many other quantization techniques .

for any feel free to open PR

!Thank you

🌐Source https://youtu.be/vW30o4U9BFE?si=HvlOwNKczBHVt69J